INCH-POUND

MIL-S-70547 (AR)
AMENDMENT 6
04 October 1994
SUPERSEDING
AMENDMENT 5
06 May 1993

MILITARY SPECIFICATION

# SPHEROIDAL PROPELLANT, M38 FOR USE IN PROPELLING CHARGES M218, M219, AND XM220 FOR CARTRIDGES, 81MM, SMOKE, M819, ILLUMINATING M853 AND PRACTICE, XM879

This Amendment forms a part of Military Specification MIL-S-70547 (AR) dated 5 September 1986, and is approved for use by the U.S. Army Armament Munitions and Chemical Command and is available for use by all Departments and Agencies of the Department of Defense.

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Title: Add "(M220)" and "(M879)". Make these changes whenever reference to XM220 and XM879 appears in the specification.

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3.2.1 Chemical composition:

Table I Composition: Delete "0.85 ± 0.25" and substitute "0.85 to 1.50".

3.4 Table II, III and IV: Change Individual Chamber Pressure (Max) at 145 degrees F from "13,100" to "14,500".

Table III: Change Muzzle Velocity (Mean) at 70 degrees F from "1020 ft/sec" to "1013 ft/sec".



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4.4.1.3 Delete in its entirety and substitute new paragraph 4.4.1.3:

"4.4.1.3 Ballistic performance. Five containers shall be selected at random from each lot of propellant. Equivalent quantities of the five containers shall be taken such that the aggregate quantity shall be sufficient for ballistic performance tests described in 4.5.3. The propellants shall be aged a minimum of forty-five (45) days at sealed and ambient conditions from the date of the packing (unless other conditions specified), prior to ballistic testing."

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5.1.3 Level C. Add new paragraph as follows:

"In addition, the following requirements apply: All metal parts shall be hot dipped galvanized. Top and bottom chime shall be 24 gauge steel and shall be welded. The body shall be wound with a hot melt or thermo plastic adhesive. The bottom shall be waterproof laminated fiber board. The body and bottom disc shall also have a laminated aluminum foil barrier. The bottom crimp shall be caulked. The finished drum with closure assembled shall be moisture proof and leak tight."

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* 5.2.3 Delete in its entirety and substitute new paragraph 5.2.3:

"5.2.3 Level C. Marking shall be in accordance with Title 49 CFR Section 173.93 and MIL-STD-129. UN proper shipping name and UN Identification Number shall be Powder, Smokeless UN 0161."

The margins of this amendment are marked with an asterisk or vertical lines to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous amendment.

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